Project proposal

By Ioan Cristian Bratu, Marina Hornero Merino, Krisa CARKA, Youssef EL OTMANI

Project 1: Alzheimer’s disease detection

- **Objective:** Detection of Alzheimer's disease through natural language processing (NLP) techniques applied to speech recognition data. We want to improve previous benchmarks (84% accuracy) https://github.com/42bismuth/Alzheimer-Detection?tab=readme-ov-file#dataset
- **Dataset**: https://talkbank.org/dementia/ Contains audios of 117 people with Alzheimer’s and 93 healthy individuals.
- **Models:** neural networks
- **Split:** ‘train’: 393, ‘test’: 102, ‘validation’: 57

Project 2 : Multi-Turn "Instruction Extraction"

- **Objective** : reconcile information across threads. (Cross-Email Information Extraction and Context Retrieval)
  - **Step A:** Single-Turn Extraction: Identify potential tasks in each email.
  - **Step B:** Thread Resolution: If Email 1 says "Can you send the report?" and Email 3 says "I'll do it by Friday," the goal is to output a single object:
    { "Task": "Send report", "Assignee": "Sender of Email 3", "Deadline": "Friday" }
- **Dataset :** MailEx (A subset of Enron consisting of ~1,500 email threads specifically annotated for Event and Argument Extraction.)
- salokr/Email-Event-Extraction
- **Models :** For example : Longformer , all-MiniLM-L6-v2